Yinxi Li (Larry)

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Education

University of Waterloo, Canada *May 2025 – April 2027*
M.Math. in Computer Science, GPA: 97.7/100, Advisor: Pengyu Nie *(Expected)*
The Chinese University of Hong Kong, Hong Kong (SAR) *Sept 2020 – July 2024*
B.Sc. in Computer Science, ELITE Stream
ETH Zürich, Switzerland *Feb 2023 – Sept 2023*
D-INFK, Exchange Student

Research Interests

Agents	LLM agents and test-time learning: self-evolving agents, agent memory and computer-use agents
Reasoning	Reasoning: latent and looped reasoning, agentic search, code language models, test generation, reasoning models for protein design, agents for scientific software
Efficiency	Efficient inference and context-to-weights: KV-cache compression, hypernetwork-generated adapters
Architecture	LLM architecture: quantized latent states, looped architecture LLMs, tokenization

Publications and Manuscripts

Journal Articles and Conference/Workshop Papers

Yinxi Li*, Qingcheng Zeng*, Weihao Xuan, Heli Qi, Fuheng Zhao, Puxuan Yu, Kaize Ding, Pengyu Nie, and Tianyu Liu. "**FINDER: Studying and Repairing What Test-Time Learning Changes in Deep-Research Agents.**" In *Proceedings of the ACM International Conference on Web Search and Data Mining (WSDM)*, 2027. [code] Agents Reasoning

Weihao Xuan*, Qingcheng Zeng*, Jiarui Liu*, Rui Yang*, Yunze Xiao, **Yinxi Li**, Zeqi Zhou, Heli Qi, Leyang Cui, Tianhao Ma, Hanqun Cao, Junjue Wang, Tianyu Liu, Weihua Du, Yueqi Song, Jiayi Geng, Matthew Yu Heng Wong, Rob Voigt, Chuan Hong, Xiang Yue, Jen-Tse Huang, Xuhai Xu, Yifan Peng, Bernhard Schölkopf, Zhijing Jin, Pheng Ann Heng, Nan Liu, Naoto Yokoya, and Mona Diab. "**Learning from Use: Test-Time Learning in Large Language Models and Agents.**" *Nature Reviews Computing*, accepted, 2026. SSRN Agents

Yinxi Li, Yuntian Deng, and Pengyu Nie. "**TokDrift: When LLM Speaks in Subwords but Code Speaks in Grammar.**" In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026. arXiv:2510.14972 [code] Architecture

Fang Wu*, Weihao Xuan*, Heli Qi*, Hanqun Cao*, Heng-Jui Chang*, Zeqi Zhou*, Haokai Zhao, Ma Jian, Carl Ma, Yu-Chi Cheng, Kuan Pang, Xiangru Tang, Zehong Wang, Guanlue Li, Hanchen Wang, Kejun Ying, Pan Lu, Chiho Im, Seungju Han, Peng Xia, Tinson Xu, **Yinxi Li**, Deyao Zhu, Pheng-Ann Heng, Naoto Yokoya, Masashi Sugiyama, Li Erran Li, Jure Leskovec, Yejin Choi. "**Proteo-R1: Reasoning Foundation Models for De Novo Protein Design.**" In *Proceedings of the International Conference on Machine Learning (ICML)*, 2026. arXiv:2605.02937 [project] [code] Reasoning

Yinxi Li, Jiaao Chen, Fang Wu, Jiakai Yu, Heli Qi, Weihao Xuan, Haokai Zhao, Pengyu Nie, Di Jin, and Xiangru Tang. "**Learning Multi-step Reasoning via Persistent Latent State Propagation.**" In *ICLR 2026 Workshop on Latent & Implicit Thinking – Going Beyond CoT Reasoning*, 2026. Reasoning Architecture

Preprints and Manuscripts Under Review

Dingyuan Dai*, Heli Qi*, Lei Liu*, **Yinxi Li***, Baiding Chen*, Zijun Dou*, Qingcheng Zeng, Qi Kang, Kunyang Sun, Eric Wang, Bo Zhou, Haixin Wang, Yufan Du, Shi Bo, Ruihan Lin, Mengqi Yuan, Dunjie Lu, Steven Dillmann, Yiming Shi, Yunshan Guo, Minghao Liu, Ge Zhang, Pengyu Nie, Zhen Yang, Jie Tang, Juanzi Li, Weihao Xuan, and Tianyu Liu. "**OSWorld-Science: A Benchmark of Computer Use Agents for Learning and Using Scientific Software.**" *Manuscript under review.* [project] [code] Agents Reasoning

Tianyu Liu, **Yinxi Li**, and Qingcheng Zeng. "**PSPO: Preference-Shrunk Uncertainty-Aware Policy Optimization.**" *Manuscript under review.* Reasoning

Bihui Jin, **Yinxi Li**, Kaiyuan Wang, and Pengyu Nie. "**Assembling Insights for Agentic Machine Learning Engineering Systems.**" *Manuscript under review.* Agents

Liliia Hotsko, **Yinxi Li**, Yuntian Deng, and Pengyu Nie. "**Code2LoRA: Hypernetwork-Generated Adapters for Code Language Models under Software Evolution.**" *Manuscript under review.* arXiv:2606.06492 [project] Reasoning Efficiency

Aaron Tu*, Weihao Xuan*, Heli Qi*, Xu Huang*, Qingcheng Zeng, Shayan Talaei, **Yinxi Li**, Yijia Xiao, Peng Xia, Xiangru Tang, Yuchen Zhuang, Bing Hu, Hanqun Cao, Wenqi Shi, Rui Yang, Nan Liu, Huaxiu Yao, Ge Liu, Li Erran Li, Amin Saberi, Naoto Yokoya, Jure Leskovec, Yejin Choi, and Fang Wu. "**Position: The Hidden Costs and Measurement Gaps of Reinforcement Learning with Verifiable Rewards.**" *Manuscript under review at AAAI 2027.* arXiv:2509.21882 Reasoning

Fang Wu, Yunxiang MO, Heli Qi, Weihao Xuan, Tinson Xu, Hongbin Lin, Yanchao Li, Zeqi Zhou, Kejun Ying, Hanchen, Peng Xia, Xiangru Tang, Zehong Wang, Ningshan Ma, **Yinxi Li**, Jiakai Yu, James Zou, Jure Leskovec, Yejin Choi. "**Robust Information-Gain Control Active Reasoning under Approximate Beliefs.**" *Manuscript under review.* Reasoning

Fang Wu*, Haokai Zhao*, Da Xing, Hanqun Cao, Tinson Xu, Xinyu Xiang, Yanchao Li, Xiangru Tang, Zehong Wang, Kuan Pang, Hanchen, Hongbin Lin, Zeqi Zhou, **Yinxi Li**, Peng Xia, Aaron Tu, Li Erran Li, Molei Tao, Jure Leskovec, Aditya Joshi, Yejin Choi. "**NoiseRater: Meta-Learned Noise Valuation for Diffusion Model Training.**" *Manuscript under review.* arXiv:2605.08144

Project Experience

KV Cache Eviction on Hybrid Attention Reasoning Models *Sep 2026 – Present*
Efficiency Architecture

Learning KV Cache Memory into Model Weights at Test Time *June 2026 – Present*
Efficiency Architecture

- Designing a theoretical and empirical study to investigate the effectiveness of learning KV cache memory into model weights at test time.

Self-Evolving Harness for Computer-Use Agents on Scientific Software *June 2026 – Present*
Agents Reasoning Efficiency

- Developing a self-evolving agent harness that learns from interaction trajectories, execution feedback, and prior task experience to improve computer-use agents on scientific software.

Quantized Recursive Model: Advancing Looped Reasoning with Quantized Tree Search *Feb 2026 – May 2026*
[code] Reasoning Architecture

- Extended Tiny Recursive Model with a stochastic finite-scalar-quantized latent bottleneck and Monte-Carlo tree search over quantized reasoning states, trained with a reward- and depth-weighted tree loss.

Unit Test Transplantation with Retrieval-Augmented Generation *Sept 2024 – Dec 2024*

- Developed a RAG pipeline for test transplantation to improve unit test generation in API frameworks.
- Analyzed model performance on 2,000+ unit test cases extracted from 7 diverse open-source repositories, identifying optimization opportunities and proposing enhancement.

Optimizing Structure of Vision-Language Retrieval System *Aug 2023 – June 2024*
ELITE Stream Final Year Project, Supervisor: Eric Lo

- Developed and integrated a proxy model and optimized retrieval structures, achieving up to 50% faster object detection and 1000× inference speedup with minimal accuracy loss.
- Designed a vision-language retrieval system for large-scale video datasets, enabling efficient and diverse top-K results with reduced computational complexity from $\mathcal{O}(n^2d)$ to $\mathcal{O}(nd)$.
- Conducted comprehensive evaluations, including baseline comparisons and ablation studies, demonstrating superior retrieval accuracy, diversity, and architectural effectiveness.

Adaptive Streaming for Short-Video Services *June 2022 – Aug 2022*
CUHK Summer Research, Supervisor: Lee Yiu-Bun

- Implemented and evaluated multiple adaptive bitrate (ABR) algorithms (including BBA, BOLA, and the RL–based Pensieve), and tuned the simulator to support focused testing on short video samples.

Awards and Scholarships

Graduate Student Research Dissemination Award, University of Waterloo *2026*
International Master's Award of Excellence, University of Waterloo *2025 – 2027*
Graduate Research Studentship, University of Waterloo *2025 – 2027*
Yasumoto International Exchange Scholarship, CUHK *2022 – 2023*
Mr & Mrs Ko Scholarship, Wu Yee Sun College, CUHK *2022 – 2023*

Competition

World University Students' Digital Intelligence Competition Invitational Tournament (by Tencent) *April 2023 - July 2023*
Ranked first among overseas institutions, Mentor: Tan LEE

- Developed a multi-agent reinforcement learning framework for Honor of Kings, with optimized reward functions and customized agent strategies.

Research Internship Experience

Research Assistant *Remote*
Southern University of Science and Technology, Supervisor: SONG Xuan *Aug 2024 - Oct 2024*

- Developed an expert knowledge collaboration system for the automotive quality inspection value chain based on large language model and retrieval-augmented generation.

Research Intern *Shenzhen, China*
Research Institute of Tsinghua University in Shenzhen *April 2024 - June 2024*

- Utilized a variational autoencoder for anomaly detection, improving customer load monitoring.

Research Intern *Changchun, China*
Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences *June 2020 - Aug 2020*

Research Community Service

Teaching Assistant: CS 246, CS346, CS446, CS486 (25 Spring, Fall, 26 Winter, Spring, Fall UWaterloo)

Sub-Reviews: ASE 2025, ARR Oct 2025

Association Activities Experience

CUHK Mainland Undergraduate Association (19th & 20th cabinet) *Aug 2021 - July 2023*

- Principal of Information Department and Assistant of Sports Department

CUHK Mainland Undergraduate Association Orientation Camp *Aug 2021 - July 2023*

- Group Leader(2021-23): Mentored 7 groups and supported 60+ freshmen in academic and campus life

Additional Information

Open-Source Codebase Experience: Hands-on experience using, extending, and modifying LLM inference, agent harness, model-training and informational retrieval codebases, including vLLM, SGLang, LangGraph, Hugging Face Transformers, Accelerate, Tokenizers, Megatron-LM, verl, FAISS, and Tevatron.

Selected Coursework: Machine Learning for SE, Computational Linguistics, Mining & Analysis of SE Data, Foundations of Optimization, Machine Learning Theory, Fundamentals of AI, Rigorous Software Engineering, Introduction to ML, Data Modelling and Databases

Programming Languages: Python, Java, Shell Script